Hybrid Biomedical Image-Analysis Pipeline: Cell Nuclei

GITHUB Link:https://github.com/IswaryaLaharikola/AI-Imaging-Assignment-3

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1. Overview and Data Preparation

This work introduces an end-to-end biomedical image processing pipeline on cell nuclei images using Qwen2.5-VL-3B-Instruct, traditional image processing using scikit-image, and a lightweight PyTorch U-Net. The pipeline includes image → segmentation → quantitative features → structured JSON → narrative, blending both visual understanding and quantitative measurement of image features.

The dataset comprised 80 training images/masks, 20 validation images/masks, and 12 novel test images. The images were first converted to grayscale and resized to 256 × 256 pixels. From exploratory analysis, the images had a dark background with a brighter foreground. This justified the use of Otsu thresholding as a transparent benchmark, while indicating possible issues arising from object overlap and intensity variations.

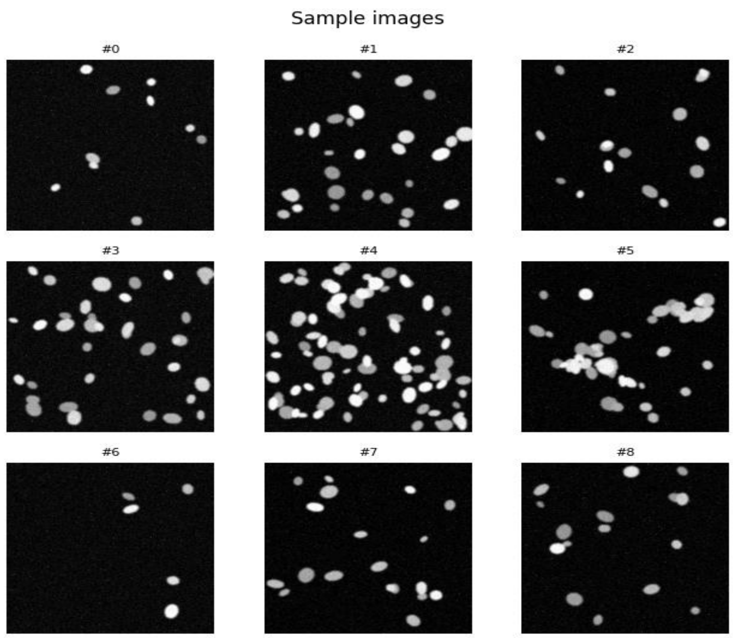

Figure 1. Sample preprocessed training images.

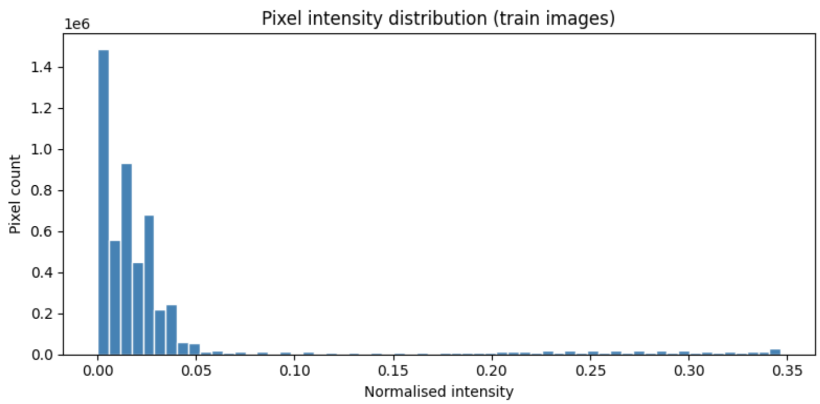

Figure 2. Pooled training-image intensity histogram.

2. Task 1 — Data Preparation, EDA and Multimodal LLM Description

The images were converted into greyscale and resized to 256 x 256 pixels. The data set consisted of 80 training, 20 validation, and 12 test image-mask pairs. Exploratory data analysis (EDA) involved sample images and an intensity histogram revealing a dark background and brighter foreground objects.

The Qwen2.5-VL-3B-Instruct image model was allowed as an alternative to llama3.2-vision. A naive prompt was compared to an optimised prompt which constrained the model to descriptive instead of diagnostic analysis. The optimised prompt had the required fields modality, tissue_type, notable_features, and image_quality, where "uncertain" was possible if evidence was lacking.

3. Task 2 — Classical features and LLM interpretation.

Otsu thresholding was used along with morphological cleanup on the sample image, then the connected component labelling technique was performed, and regionprops_table features extraction was carried out. It revealed that there are 7 objects, and the mean area, eccentricity, solidity and mean intensity values are 215 pixels, 0.61, 0.96 and 0.22 correspondingly.

All the numeric information was provided to Qwen2.5-VL without image itself. LLM categorised this image as dense one with an irregular shape and “check” quality indicator. In comparison with the direct image description from Task 1, numbers-first approach is more quantitative and auditable while direct VLM provides more visual interpretations.

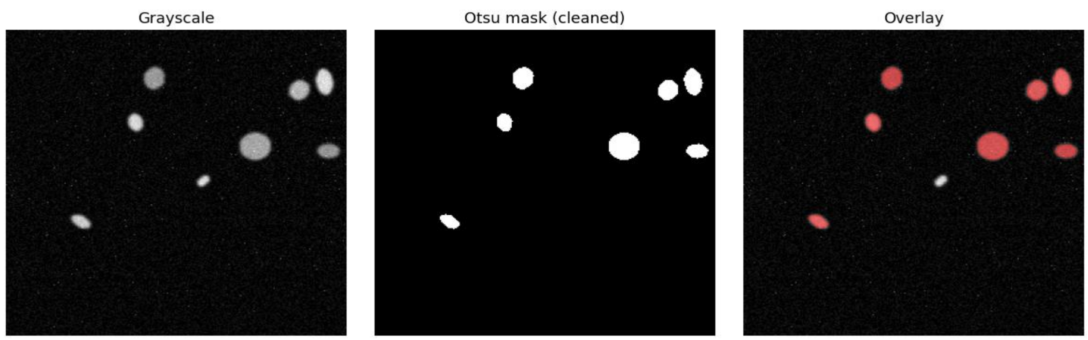

Figure 4. Grayscale Image, Cleaned Otsu Mask and Segmentation Overlay

Feature	Result	Meaning	Source
n_objects	7	Detected objects	regionprops
Area	120–428 px	Object size range	regionprops
Mean area	215 px	Average object size	regionprops
Eccentricity	0.61	Moderate elongation	regionprops
Solidity	0.96	High convexity	regionprops
Mean intensity	0.22	Low mean object intensity	regionprops

Table 1. Representative classical feature summary.

4. Task 3 — U-Net Segmentation

A simple three-layer U-Net architecture was then trained on the train data for 30 epochs. This was done with the use of an Adam optimiser with a learning rate of 1×10−3, along with a combined loss function of BCE + Dice. The validation loss, Dice and IoU were monitored at every epoch.

The resulting validation performance was Dice = 0.997 and IoU = 0.993, which shows that there is very good overlap between the predicted and ground truth masks. The training and validation plots were also studied in order to understand the training dynamics of the network. Additionally, several validation images were taken and shown alongside the ground truth mask and the corresponding U-Net output.

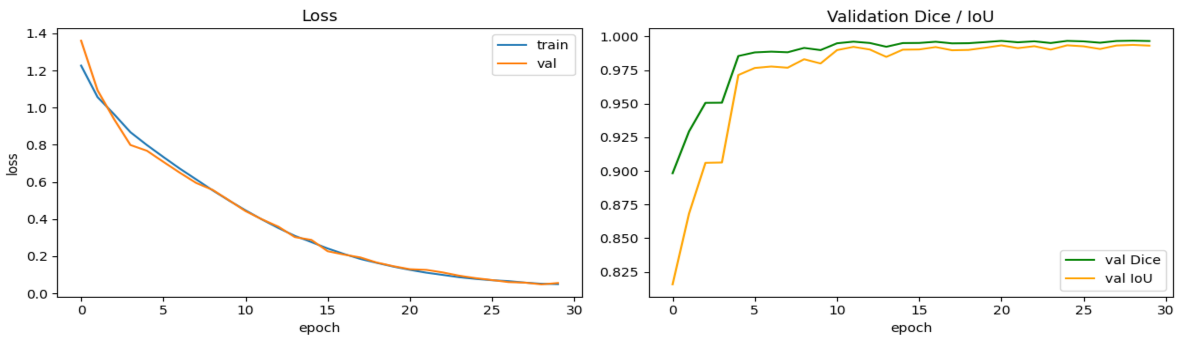

Figure 4. Training/validation loss and validation Dice/IoU curves.

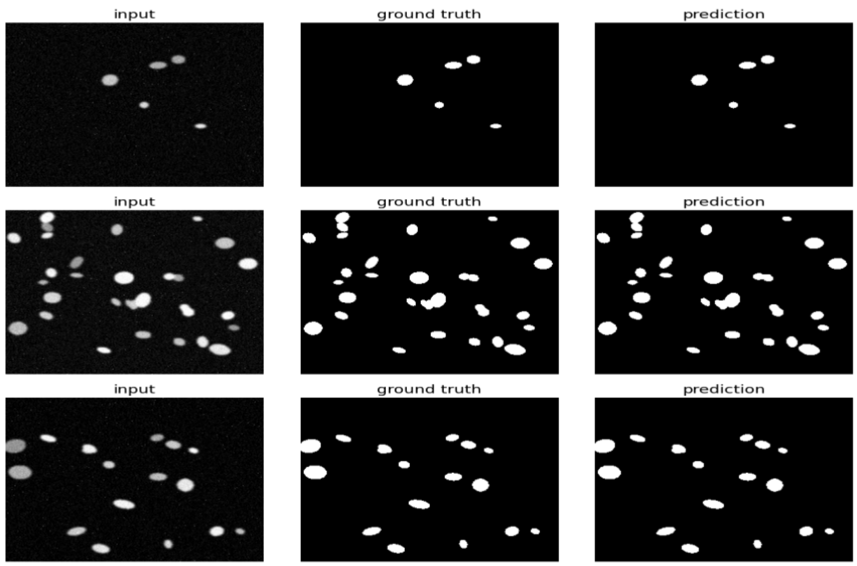

Figure 5. U-Net validation panels: input, ground truth and prediction.

5. Task 4 — Hybrid Pipeline on Unseen Test Images

The entire pipeline process was performed on the 12 unseen test images. The segmentation predictions from U-Net were mapped into region-level attributes with regionprops, and the numerical results were fed to Qwen2.5-VL without any input images. The LLM created JSON files and stories for each image.

The JSON results were combined into one pandas DataFrame file called aggregated_records.csv. It was found that there was variation in the number of objects among the test images. The numerical segmentation output was retained as the source of truth while the LLM was used solely for interpretation and story creation.

Image	n_objects	Density	Regularity	Quality
test_000	8	sparse	irregular	check
test_001	14	dense	irregular	check
test_002	26	dense	irregular	check
test_003	19	dense	irregular	check
test_004	43	dense	irregular	check
test_005	16	dense	irregular	check
test_006	8	dense	irregular	check
test_007	23	dense	irregular	check
test_008	23	dense	uncertain	check
test_009	21	dense	irregular	check
test_010	38	dense	irregular	check
test_011	20	dense	irregular	check

Table 3. Aggregated hybrid JSON fields for the 12 unseen test images.

6. Critical Analysis — Required Questions

Q1. Which description is more useful, and which is more trustworthy, the direct VLM description (Task 1) or the numbers-first description (Task 2) and why?

The description that goes first with the numbers is more reliable since the LLM receives only the quantitative features extracted from the segmentation and not the image itself. Hence, the output is limited to measurable data. The description which goes directly with the VLM is more useful as it helps interpret the visual content, but it has shown inconsistencies in terms of uncertainty and variability – the naive prompt received "uncertain" response for all fields, whereas the optimised one pointed to bright rounded objects on a dark background. Five repetitions resulted in 5/5 unique outputs, which means variability. Hence, the numbers-first description gives more auditability, while the direct VLM gives rich qualitative content.

Q2. Did the U-Net improve on classical Otsu segmentation for your modality? Give one example image where each approach did better.

Yes. The notebook has done a direct comparison on the same validation images. The mean Dice is 0.997 for the U-Net against 0.972 for the Otsu, which means that the learned segmentation is better than the classical Otsu one.

The notebook does not give any data about the per-image Dice values or specify any image IDs that did better with Otsu or U-Net, hence no specific example image should be provided if you do not see it in the validation comparison plots you saved.

Q3. Report your U-Net's Dice and IoU. What do these numbers mean, and where does the model tend to make its mistakes (which images or regions)?

The Dice and IoU scores were respectively 0.997 and 0.993. During training, Dice and IoU evolved as follows: epoch 1: Dice 0.898 / IoU 0.816; epoch 30: Dice 0.997 / IoU 0.993.

These values are indicative of a high level of agreement between the predicted and ground-truth masks. The notebook gives three input-ground truth-prediction validation panels for visual inspection, however, it does not give a numerical per-image error breakdown and particular images/regions causing any errors. Consequently, the report should use visual predictions panels rather than trying to give a concrete breakdown that is not captured by the notebook.

Q4. Where in the pipeline can the LLM hallucinate, and what design choices reduce that risk? Why does keeping the structured JSON as the "source of truth" help?

LLM can hallucinate if the LLM interprets an image directly or creates a narrative on the image going beyond the available evidence. Hallucination is prevented by using the prompt with the descriptive rather than diagnostic nature, allowing for "uncertainty," forcing a certain JSON structure, and having a numbers-first phase where LLM gets numerical measurements instead of the image itself.

Having structured JSON as the source of truth increases the audibility of the pipeline because the narrative can be checked against the underlying numerical measurements..

Q5. Considering accuracy, auditability, and the limits of your dataset, would you trust any part of this system in a real clinical setting? What single change would most improve trustworthiness?

No, Not for any clinical decision making process in the current state. The notebook managed to get excellent results when it comes to validation segmentation; however, it is an educational pipeline and the output from LLM shows stochastics. Additionally, the notebook itself states that the system is not clinically cleared and hallucinations are the limitations.

The most important step for improvement is external validation with a different set of data, which includes expert labeling.

7.Overall Interpretation and Design Trade-offs

In this design choice, the pipeline involves the use of segmentation techniques from classical approaches, U-Net and the language models for interpretations. The output in the form of JSON files ensures consistency, whereas the features in numerical form give an audit trail for the description by the language model.U-Net has performed well on validation, but there is limited data to ensure generalization. Hence, the mixed method takes the output of segmentation quantitatively as the ground truth.

8. Optimised Prompts Used

Task 1 – Direct image interpretation: The instructions given to Qwen2.5-VL were to only describe the visual aspects of the image, no diagnosis, use “uncertain” where information could not be provided, and provide the modality, tissue_type, notable_features, and image_quality.

Task 2 – Quantitative-first interpretation: Only the numerical features of the segmented image were given to the model. The instruction was for the model to interpret these numerical data points without seeing the image and to provide the n_objects, density_class, shape_regularity, and quality_flag.

Task 4 – Hybrid interpretation: The LLM interpreted the numerical data generated using the U-Net segmentation and produced structured notes along with image_id, n_objects, mean_area, density_class, and quality_flag.

9. References

- Ronneberger, O., Fischer, P. and Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation*. In: Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 234–241.
- Otsu, N. (1979). *A Threshold Selection Method from Gray-Level Histograms*. IEEE Transactions on Systems, Man, and Cybernetics, 9(1), pp. 62–66.
- van der Walt, S. et al. (2014). *scikit-image: Image processing in Python*. PeerJ, 2, e453.
- Paszke, A. et al. (2019). *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. In: Advances in Neural Information Processing Systems, 32.
- Bai, J. et al. (2025). *Qwen2.5-VL Technical Report*. arXiv preprint.

10. Conclusion

This assignment created a complete biomedical image analysis pipeline including segmentation, U-Net, and LLM interpretation. The U-Net showed excellent validation performance (Dice = 0.997, IoU = 0.993), and outperformed Otsu segmentation.

The prompts helped to structure the LLM outputs, and the pipeline helped to link segmentation outputs with quantitative metrics and LLM interpretation. Nevertheless, the small amount of data used and different answers from the LLM limit generalizability and clinical applicability.

Overall, this assignment shows the possibility to combine computer vision techniques with LLMs in biomedical image analysis. The key improvement in terms of increasing trustworthiness is external validation on a large dataset annotated by experts.